ECE 204 Design Project

8 Bit ALU

Nolan Kessler, Maxwell Fabian, John O'Connor

June 5, 2026

1 8 Bit ALU Functional Description

This 8 bit ALU accepts two 8 bit operands A and B, along with a 3 bit opcode (OP). The state of the ALU is controlled by an active high enable pin (EN), and can be reset using the active low reset pin (RST_N). As this ALU supports sequential operations which may take more than one clock cycle, the CLK input must be connected to a clock signal. The ALU provides one 8 bit result output (Y) an overflow flag (OVER) and a done flag (DONE). A table of the ALU's supported operations is given in table 1. A diagram of the inputs and outputs is shown in figure 1. A descriton of the function of each bus/pin is given in table 2.

Operation	**Opcode**	**Clock Cycles**	**Overflow Condition**	**Description**
A pass	0x0	1	N/A	Passes A to Y unmodified
B pass	0x1	1	N/A	Passes B to Y unmodified
Bitwise AND	0x2	1	N/A	Y is bitwise AND combination of operands A and B
Bitwise OR	0x3	1	N/A	Y is bitwise OR combination of operands A and B
Bitwise XOR	0x4	1	N/A	Y is bitwise XOR combination of operands A and B
Addition	0x5	1	$A + B > 255$	Y is the sum of operands A and B
Subtraction	0x6	1	$A + (\bar{B} + 1) > 255$	Y is the sum of A and the 2's complement inverse of B
Multiplication	0x7	Value of $B + 1$	$A * B > 255$	Y is the product of A and B

Table 1: Table of supported operations

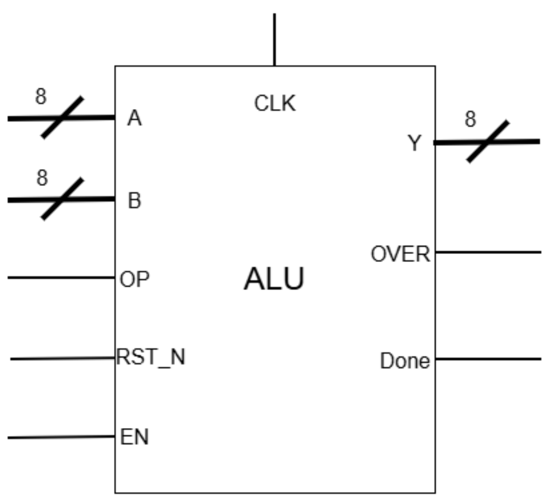

Figure 1: ALU Inputs and Outputs

Bus / Pin	Width (Bits)	Function
Input		
A	8	First 8-Bit operand - input to mathematical or logical operation.
B	8	Second 8-Bit operand - input to mathematical or logical operation.
OP	3	Opcode - selects which operation to perform.
RST_N	1	Active low asynchronous reset - puts the ALU into initial ready state.
EN	1	Enable - starts ALU when high if in ready state. Must be high for ALU to output data.
Output		
Y	8	8 Bit result - output of the ALU operation
OVER	1	Overflow flag - high value generally indicates result cannot be represented in bits of Y. Exact meaning can vary between operations.
DONE	1	Is low while operation is in progress or ALU is ready, high when operation is completed.

Table 2: ALU IO Functional Descriptions

Figure 2: ALU Operation Timing Diagram

1.1 ALU Operation

Before beginning an operation the ALU must be reset by setting RST_N low, after reset occurs the ALU is considered to be in the ”ready” state. EN must be low for all rising edges of CLK while the ALU is in the ready state. Once the inputs on A, B, and OP are stable, EN can be set high to begin ALU operation. The rising edge of EN must occur when CLK is low. EN must remain high until after the next falling edge of CLK. At this point the ALU enters the ”running” state. From this point on the value of EN can be changed without effecting the ALU (or the result when the ALU is done). The ALU will always output all zeros when EN is low. When DONE goes high the ALU is in the ”done” state. The operation is finished, and results can be read from Y, and OVER. A timing diagram of a normal ALU operation is shown in figure 2.

2 Design and Implementation

This ALU is designed to utilize modular operations, to allow the design to allow for easy testing during the design process, and potential rapid addition of new functions in the future. These modular operations are then linked together with integrated together with structures to allow for the selection of operations, forming the complete ALU.

Bus / Pin	Width (Bits)	Function
Input		
A	8	First 8-Bit operand - input to operation.
B	8	Second 8-Bit operand - input to operation.
RST_N	1	Active low asynchronous reset - resets the module to inial ready state.
EN	1	Enable - enables the module. The module will only output data when this pin is high
Output		
Y	8	8 Bit result - output of the operation
OVER	1	Overflow flag - high value generally indicates result cannot be represented in bits of Y. Exact meaning can vary between operations.
DONE	1	Is low while operation is in progress or is ready, high when operation is completed.

Table 3: Operation Module IO Descriptions

2.1 Operation Modules

Each operation module accepts the inputs, and supplies the outputs shown in table 3. All outputs of the modules are routed through tristate buffers - the outputs of an operation are floating unless its enable pin is high.

There are two primary types of operation modules utilized in this ALU, combinational logic operations, and sequential logic operations. For combinational logic operations, the CLK and RST_N inputs are not needed, and DONE is always high when the module is enabled. A diagram of such an operation module is shown in figure 3. Sequential logic operations utilize all inputs to perform operations that require more than one clock cycle. The only such operation in this ALU is multiplication. A diagram of this type of module is shown in 4.

3 Design Verification

4 Hardware Validation

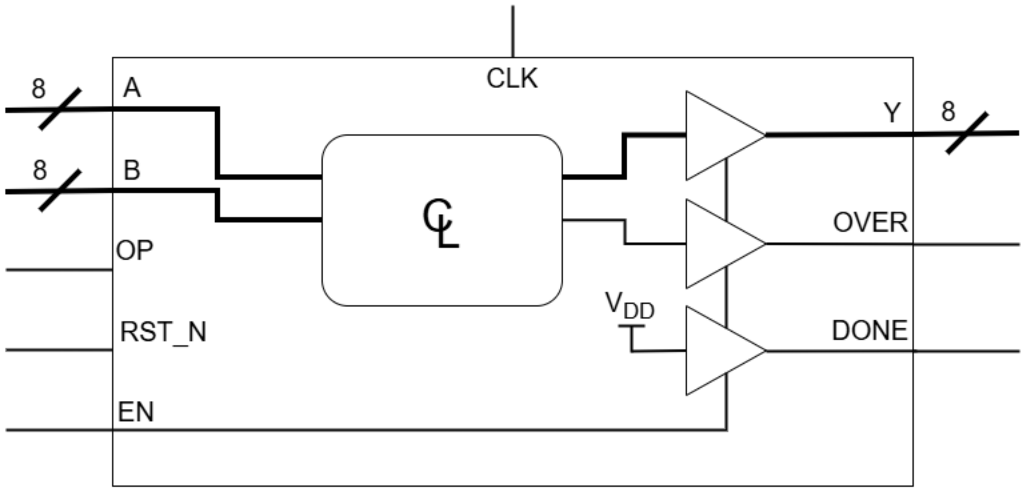

Figure 3: Schematic of a Combinational Operation Module

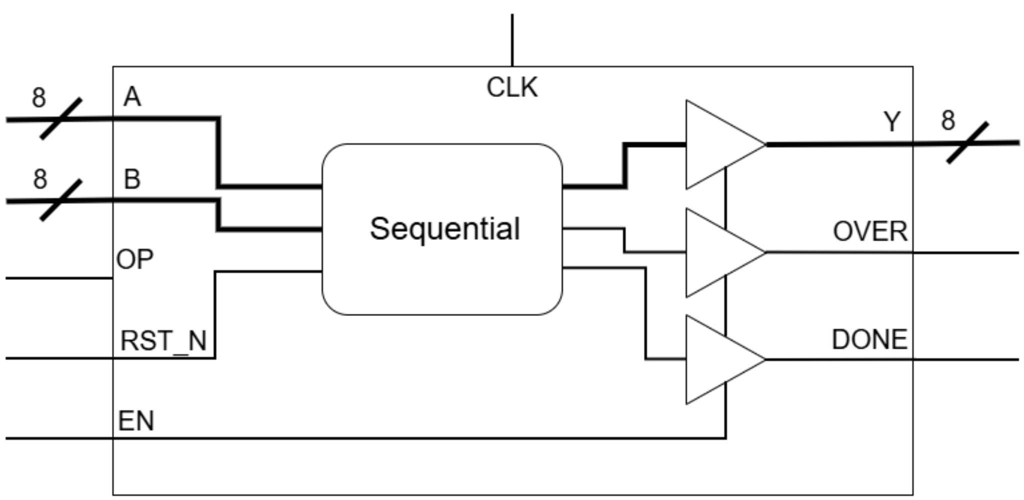

Figure 4: Schematic of a Sequential Operation Module